Version Spaces and Candidate Elimination

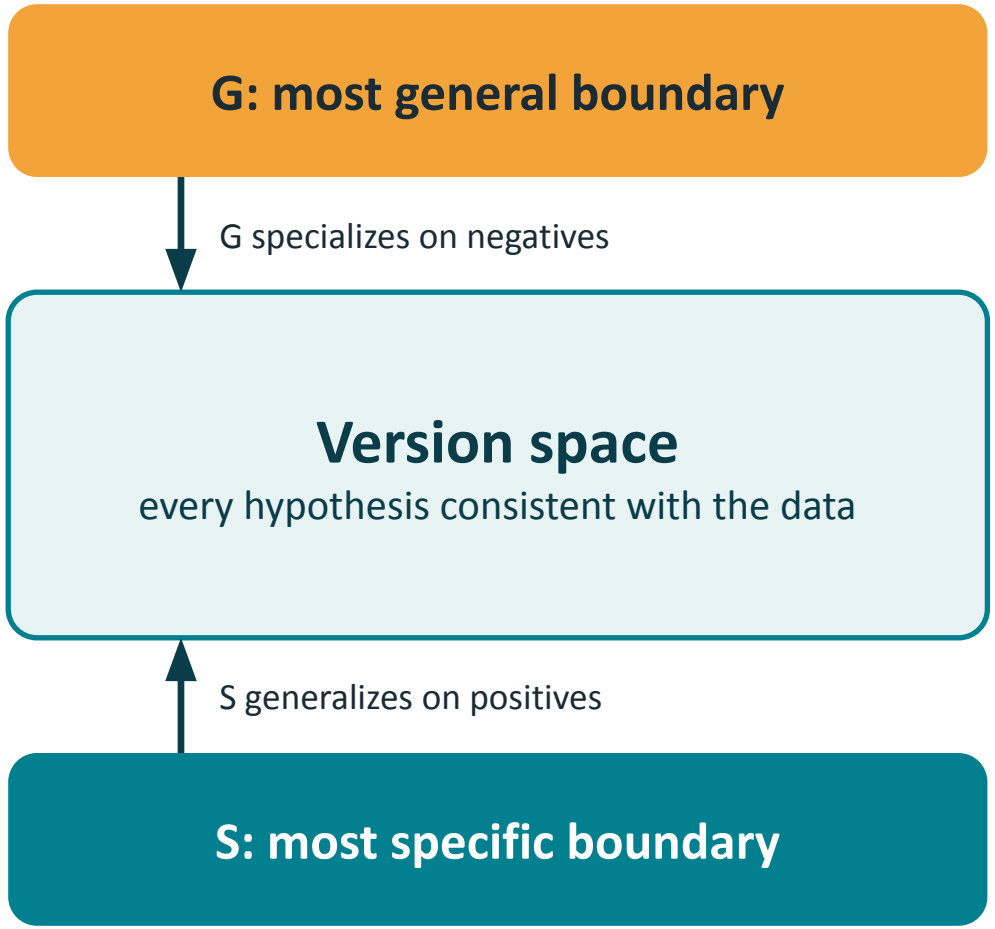
Machine Learning - ITA0611

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What is a Version Space?

- The set of all hypotheses consistent with the training examples
- Sits between the most specific and the most general hypotheses
- Shrinks as more examples arrive
- Stored compactly as two boundaries, S and G



Candidate Elimination Algorithm

1

Initialize

S = most specific hypothesis, G = most general hypothesis

2

Read an example

Take the next training example, positive or negative

3

Update S and G

A positive generalizes S; a negative specializes G

4

Repeat

Stop at S = G (concept learned) or an empty boundary (inconsistent data)

Each example can only remove hypotheses, so the version space never grows

Roles of the S and G Boundaries

S Specific boundary (S)

- Most specific hypotheses that cover all positive examples
- Moves up (generalizes) on positives

Initial: S = ⟨∅, ∅, …, ∅⟩

G General boundary (G)

- Most general hypotheses that exclude all negative examples
- Moves down (specializes) on negatives

Initial: G = ⟨?, ?, …, ?⟩

Every consistent hypothesis lies between S and G

Updating the Boundaries

+ Positive example

1. Remove from G any hypothesis that fails to cover it
2. Minimally generalize each s in S that fails to cover it
3. Keep only those still more specific than a member of G

Result: S moves up (more general)

− Negative example

1. Remove from S any hypothesis that wrongly covers it
2. Minimally specialize each g in G that covers it
3. Keep only those still more general than a member of S

Result: G moves down (more specific)

Worked Example: EnjoySport

Training example	Label	S after	G after
Start		⟨∅, ∅, ∅, ∅, ∅, ∅⟩	⟨?, ?, ?, ?, ?, ?⟩
⟨Sunny, Warm, Normal, Strong, Warm, Same⟩	+	⟨Sunny, Warm, Normal, Strong, Warm, Same⟩	⟨?, ?, ?, ?, ?, ?⟩
⟨Sunny, Warm, High, Strong, Warm, Same⟩	+	⟨Sunny, Warm, ?, Strong, Warm, Same⟩	⟨?, ?, ?, ?, ?, ?⟩
⟨Rainy, Cold, High, Strong, Warm, Change⟩	−	⟨Sunny, Warm, ?, Strong, Warm, Same⟩ (unchanged)	⟨Sunny, ?, ?, ?, ?, ?⟩ ⟨?, Warm, ?, ?, ?, ?⟩ ⟨?, ?, ?, ?, ?, Same⟩

Attributes: Sky, Temp, Humidity, Wind, Water, Forecast. The negative example specializes G just enough to exclude it.

Key Takeaways

1. The version space holds every hypothesis consistent with the data
2. S bounds it from below (specific) and G bounds it from above (general)
3. Positives generalize S and prune G; negatives specialize G and prune S
4. S = G means the concept is learned; an empty S or G means inconsistent data or a limited hypothesis space